

## Week 6 Discussion: Forwards, Futures & Swaps (Lec 14–16)

*Main focus: (i) Pricing vs. valuing a forward; (ii) futures pricing with income/costs (Master Formula); (iii) designing an interest rate swap via comparative advantage.*

### 1 Part 1: Value of a Forward Contract

#### Module A) Pricing ( $F_0$ ) vs. Valuation ( $f_t$ )

##### Crucial Distinction:

- **Forward Price ( $F_0$ ):** the delivery price  $K$  is set at inception so that the contract value is 0.
- **Value ( $f_t$ ):** the mark-to-market value of an *existing* contract at time  $t$ .

##### Non-dividend-paying stock (continuous compounding):

$$F_0 = S_0 e^{rT}.$$

##### Value of an existing long forward at time $t$ :

$$f_t = S_t - K e^{-r(T-t)} \quad \text{and} \quad f_t = (F_t - K) e^{-r(T-t)}, \quad F_t = S_t e^{r(T-t)}.$$

#### Practice A: (Textbook 5.7) Forward price and value

A 1-year long forward contract on a non-dividend-paying stock is entered into when the stock price is  $S_0 = \$40$  and the risk-free rate is  $r = 5\%$  per annum with continuous compounding.

- What are the **forward price** ( $F_0$ ) and the **initial value** of the forward contract?
- Six months later, the stock price is  $S_{0.5} = \$45$  and the risk-free rate is still 5%. What are the **forward price** for delivery in 6 months and the **value** of the original forward contract?

#### TA Key for Practice A

(a)  $T = 1$ :

$$F_0 = 40 e^{0.05} \approx 42.05, \quad f_0 = 0.$$

(b) At  $t = 0.5$ , remaining time  $T - t = 0.5$ :

$$F_{0.5} = S_{0.5} e^{r(0.5)} = 45 e^{0.025} \approx 46.14.$$

Original delivery price is  $K = F_0 \approx 42.05$ . Value of the original long forward:

$$f_{0.5} = S_{0.5} - K e^{-r(0.5)} = 45 - 42.05 e^{-0.025} \approx 3.99.$$

## 2 Part 2: Pricing Forwards with Income / Costs / Currencies

### Module B) The “Master Formula” (Cost of Carry)

The No-Arbitrage Forward Price  $F_0$  relates to Spot Price  $S_0$  via:

$$F_0 = S_0 e^{(r-q+u-y)T} \quad \text{or} \quad F_0 = (S_0 - I + U)e^{rT}$$

- **Interest ( $r$ ):** Cost of financing. **Push  $F$  Up.**
  - **Income ( $q$  or  $I$ ):** Dividends/Coupons. Holder of forward *misses* these. **Push  $F$  Down.**
  - **Storage ( $u$  or  $U$ ):** Cost to hold physical asset. Holder of forward *avoids* these. **Push  $F$  Up.**
  - **Convenience Yield ( $y$ ):** Benefit of holding physical asset. **Push  $F$  Down.**
- Special Case: Currencies (Interest Rate Parity)** Treat foreign risk-free rate  $r_f$  as a dividend yield  $q$ .

$$F_0 = S_0 e^{(r-r_f)T} \quad \text{where } r = r_{\text{domestic}}, r_f = r_{\text{foreign}}.$$

### Practice B: (Textbook 5.10) Index futures mispricing

Suppose the risk-free interest rate is 6% per annum with continuous compounding and the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405.

- What is the **theoretical** no-arbitrage futures price? (Hint: Index has yield  $q$ , no storage/convenience).
- What arbitrage opportunity does this create? (State the trades at  $t = 0$  and what happens at maturity.)

### TA Key for Practice B

$T = 4/12 = 1/3$ . Using Master Formula with  $q = 0.04, u = 0, y = 0$ :

$$F_0^* = 400 e^{(0.06-0.04) \cdot (1/3)} = 400 e^{0.006666...} \approx 402.68.$$

Market futures price  $405 > 402.68 \Rightarrow$  **futures overpriced.**

**Cash-and-carry (one index unit):**

- At  $t = 0$ : borrow \$400 at  $r = 6\%$  and buy the index; short the futures at 405.
- Over life: receive dividends at yield  $q = 4\%$  (reduces effective carry cost).
- At  $T$ : close the index position and settle the short futures (cash-settled). Net locked-in profit (at  $T$ ) is

$$405 - 400e^{(r-q)T} \approx 405 - 402.68 = 2.32 \text{ (per index unit).}$$

## 3 Part 3: Interest Rate Swaps

### Module C) Designing Swaps (Comparative Advantage + Bank Spread)

**Setup:** Two firms have different borrowing costs in fixed vs floating markets. A swap can create a *total gain*:

$$\text{Gain} = (\text{Fixed spread}) - (\text{Floating spread}).$$

- The stronger-credit firm typically has a bigger advantage in the **fixed** market.
- If a bank intermediates, it takes a spread; the remainder is split between the two firms (often equally if stated).
- Design via **target effective rates** (what each firm ends up paying) and then back out the swap fixed rates.

### Practice C: (Textbook 7.1) Swap design with bank spread

Companies A and B have been offered the following rates per annum on a \$20 million five-year loan:

|           | Fixed rate | Floating rate |
|-----------|------------|---------------|
| Company A | 5.0%       | SOFR + 0.1%   |
| Company B | 6.4%       | SOFR + 0.6%   |

Company A requires a floating-rate loan; Company B requires a fixed-rate loan. Design a swap that will net a bank (acting as intermediary) **0.1% per annum** and that will appear **equally attractive** to both companies.

- Compute the **total gain** available before the bank spread.
- If the bank takes 0.1% and the rest is split equally, what **effective** borrowing rates should A and B end up with?
- Give one concrete swap with the bank: specify who borrows what externally and the swap cashflows (fixed rates paid/received, floating legs).

#### TA Key for Practice C

- (a) Fixed spread:  $6.4 - 5.0 = 1.4\%$ . Floating spread:  $(0.6 - 0.1) = 0.5\%$ .

$$\text{Total gain} = 1.4\% - 0.5\% = 0.9\%.$$

- (b) Bank takes 0.1%  $\Rightarrow$  remaining 0.8%, split equally  $\Rightarrow$  0.4% benefit to each firm.

- A wants floating. Its direct floating quote is SOFR+0.1%. With a 0.4% improvement:

$$\text{A target} = \text{SOFR} - 0.3\%.$$

- B wants fixed. Its direct fixed quote is 6.4%. With a 0.4% improvement:

$$\text{B target} = 6.0\% \text{ fixed}.$$

- (c) One valid structure (rates per annum, notional \$20M, reset frequency consistent across legs):

- External borrowing:** A borrows fixed at 5.0%. B borrows floating at SOFR+0.6%.
- Swap through bank:**
  - A **pays** SOFR to bank and **receives** 5.3% fixed from bank.
  - B **pays** 5.4% fixed to bank and **receives** SOFR from bank.

#### Check net costs:

- A: pays 5.0% (loan) + pays SOFR (swap) – receives 5.3% (swap) = SOFR – 0.3%.
- B: pays (SOFR+0.6%) (loan) + pays 5.4% (swap) – receives SOFR (swap) = 6.0% fixed.
- Bank: receives 5.4% fixed from B and pays 5.3% fixed to A  $\Rightarrow$  nets 0.1% per annum; floating legs net to 0.